SKILL BUILDERS · UNIT 6 · SCORE 3

Functions: Practice Set

NC.8.F.1–5 · NC.8.EE.8

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.F.1

$f(x) = 4x + 1$. Find $f(2)$.

Answer: 9

1. $f(2) = 4(2) + 1$
2. $= 8 + 1 = 9$

2 Score 3 NC.8.F.1

Which table shows a function?

(A)

x	1	1	2	3
y	4	5	6	7

(B)

x	2	3	2	4
y	1	2	3	4

(C)

x	1	2	3	4
y	5	5	6	6

3 Score 3 NC.8.F.4

Find the slope from **(1, 3)** to **(4, 12)**.

Answer: 3

1. rise = 12 - 3 = 9
2. run = 4 - 1 = 3
3. slope = 9 ÷ 3 = 3

4 Score 3 NC.8.F.4

Find the slope from **(0, 6)** to **(3, 0)**.

Answer: -2

1. rise = 0 - 6 = -6
2. run = 3 - 0 = 3
3. slope = -6 ÷ 3 = -2

5 Score 3 NC.8.F.4

Which equation matches the table?

x	0	1	2
y	5	8	11

Ⓐ y = 3x

Ⓑ **y = 3x + 5**

Ⓒ y = 5x + 3

Ⓓ y = x + 5

1. `m = 3 (y goes up 3 each step)`
2. `b = 5 (y when x = 0)`
3. `y = 3x + 5`

6 Score 3 NC.8.F.2

A savings account has `y = 5x + 20` dollars after x weeks. What is the **start value**?

Answer: **20dollars**

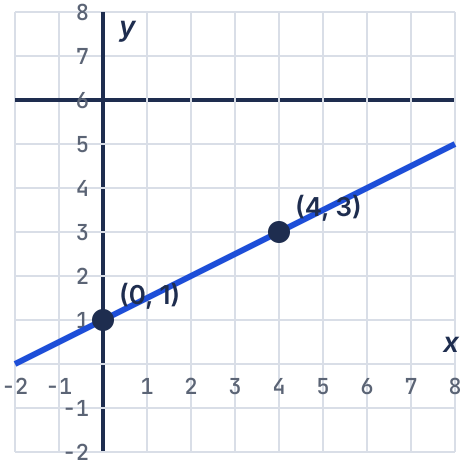
7 Score 3 NC.8.F.2

In `y = 5x + 20`, what is the **rate** (amount added each week)?

Answer: **5dollars**

8 Score 3 NC.8.F.4

What is the slope of this line?



Answer: **0.5**

1. `rise = 3 - 1 = 2`
2. `run = 4 - 0 = 4`
3. `slope = 2 ÷ 4 = 1/2`

9 Score 3 NC.8.EE.8

Solve the system: **y = x + 2** and **y = 3x**.

A **(1, 3)**

Ⓑ (2, 4)

Ⓒ (3, 1)

Ⓓ (0, 2)

1 $x + 2 = 3x$ ↓ Subtract x

$-x \quad -x$

2 $\frac{2}{2} = \frac{2x}{2}$ ↓ Divide by 2

3 $1 = x$

1 y = 3(1) = 3 → (1, 3)

10 Score 3 NC.8.F.5

A runner's distance-from-home graph is **flat** for a while. What is the runner doing?

A **Stopped: the distance is not changing**

Ⓑ Running faster

Ⓒ Running home

11 Score 3 NC.8.F.4

Find the mistake. For (2, 1) and (4, 7), Kai wrote: `slope = (4 - 2) ÷ (7 - 1) = 2 ÷ 6 = 1/3`. What went wrong?

A **Kai put the run on top. Slope = rise ÷ run = 6 ÷ 2 = 3.**

Ⓑ Kai is right.

Ⓒ The slope should be negative.

Ⓓ Kai should add: (4 + 2) ÷ (7 + 1).